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1. Introduction

1.1 Purpose

This document specifies the requirements for ShelfLife, a management system for a single general retail store. It refines the needs and features from the Vision Document into a use-case model (use-case diagrams and use-case specifications) and a set of testable requirements, following the RUP, use-case driven approach taught in the course.

1.2 Scope

ShelfLife covers the operations of one general store: inventory (from bulk purchase to unit sale), sales at manager-controlled prices, purchasing and supplier management with flexible payment terms, invoicing of customers who buy on credit, recording of operating expenses, and role-based access. Simple stock-keeping projections are included as a convenience. Out of scope for this release: multi-branch operations, multiple currencies, ERP-grade demand forecasting, and external integrations such as barcode scanning or receipt printing.

1.3 Definitions, Acronyms, and Abbreviations

Term	Meaning
SKU	Stock Keeping Unit, an identifier for a distinct product
Bulk unit	The quantity a product is purchased in (for example, a dozen)
Sale unit	The quantity a product is sold in (for example, a single item)
Unit breakdown	The fixed conversion from a bulk unit to sale units (for example, 1 dozen = 12 units)
Velocity	The historical rate at which a product sells (for example, units per day)
PO	Purchase Order
RBAC	Role-Based Access Control
COGS	Cost of Goods Sold
RUP	Rational Unified Process

2. Overall Description

2.1 Product Perspective

ShelfLife is an independent, self-contained, multi-tier web application organized into cooperating subsystems that share one database:

- Authentication and Permissions: sign-in and role-based access control.
- Inventory: products and SKUs, bulk-to-unit breakdown, stock levels, and velocity-based low-stock alerts.
- Sales: recording sales at manager-set prices and computing per-item profit.
- Purchasing and Suppliers: suppliers, purchase orders, delivery sign-off, and supplier payments.

- Customer Invoicing: credit sales and customer balances.
- Expenses and Accounting: operating expenses and financial reporting.
- Analytics and Projections: dashboards and simple stock projections.

2.2 Actors

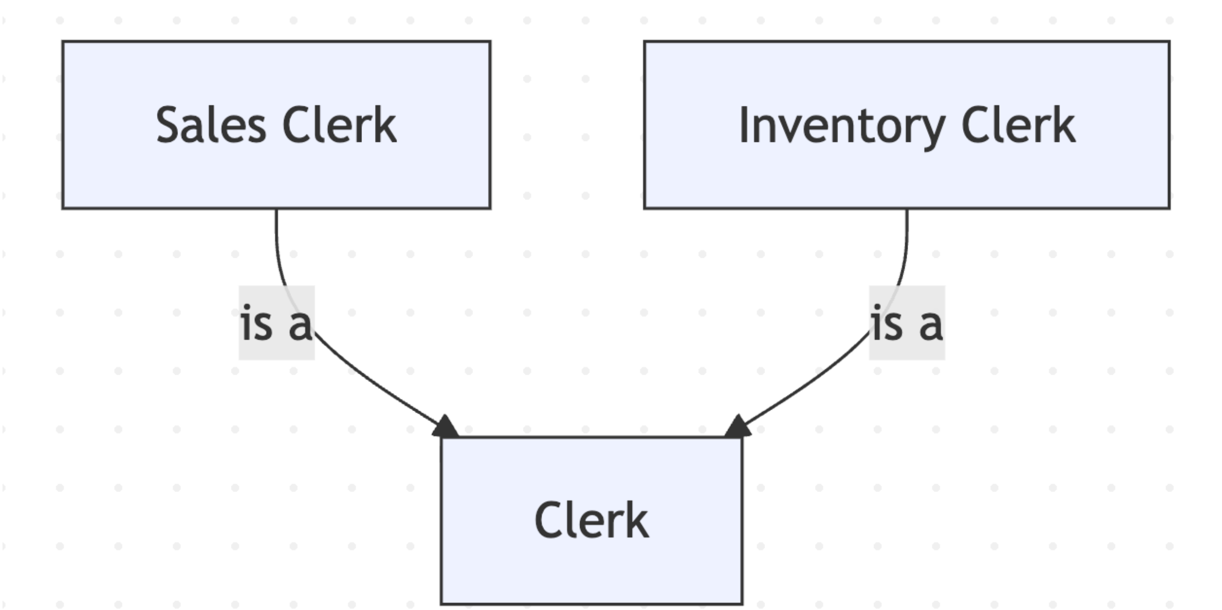
Primary (human) actors

Actor	Description
Admin / Owner	Sets up the store, holds root rights, invites employees, assigns roles, defines categories, and views all analytics.
Manager	Sets prices, creates purchase orders, approves received stock, and reviews analytics. May be the same person as Admin/Owner.
Sales Clerk	Records sales at manager-set prices and issues credit invoices.
Inventory Clerk	Adds products and stock in bulk, defines the unit breakdown, and records and signs off deliveries. Read and write only, with no editing after publish.
Accountant	Records expenses and views financial reports for tax.

Secondary (system) actors (adopting the eBazaar convention of naming back-end participants as actors)

Actor	Description
Inventory Database	Stores products, SKUs, stock levels, categories, suppliers, and purchase orders.
Accounts Database	Stores users, roles, customer and supplier balances, and expenses.
Rules Engine	Validates business rules at runtime (stock and quantity checks, unit-breakdown integers, price control, supplier-payment terms).

Actor generalization: Sales Clerk and Inventory Clerk are specializations of a general Clerk. One clerk may hold either or both responsibilities.



2.3 Assumptions and Dependencies

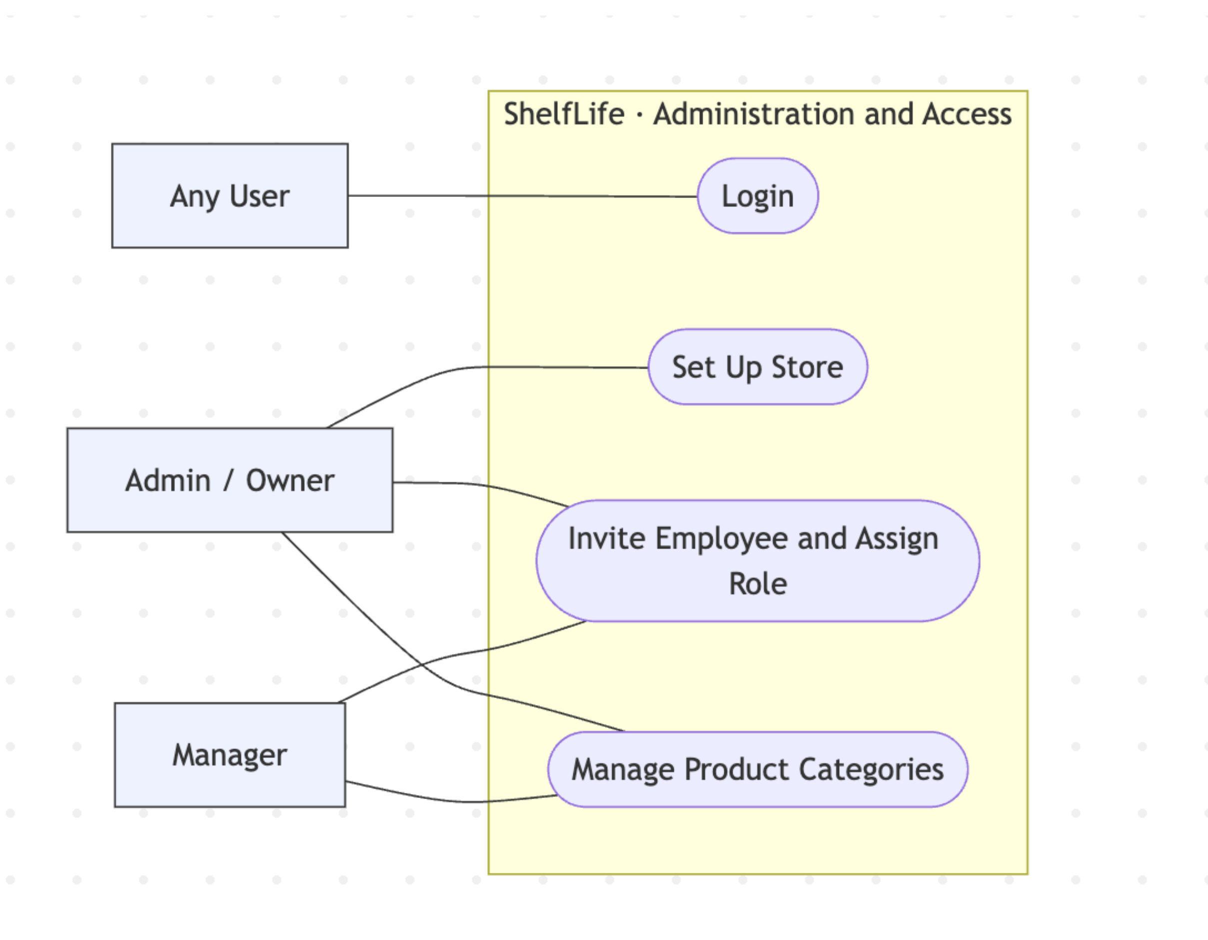
- One store, one currency, configured at setup.
- Users reach the system through a modern web browser with network access.
- The Admin/Owner performs initial setup before other roles use the system.
- Products are bought in consistent bulk units expressible as a fixed integer number of sale units.
- The implementation stack is the developer's choice (here, TypeScript on the Bun and Node.js runtime), consistent with the course's technology-neutral guidance.

3. Use-Case Model

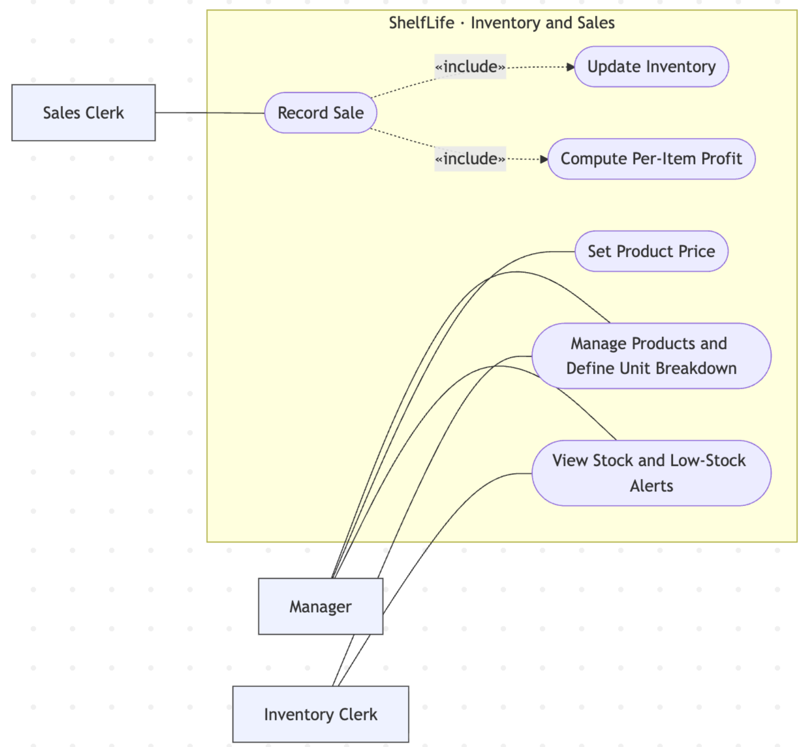
3.1 Use-Case Diagrams

Use-case diagrams are shown per subsystem for readability. Actors are shaded boxes, use cases are rounded, the box is the system boundary, and dotted arrows carry the UML stereotypes «include» and «extend». Secondary (system) actors participate in the flows as described in the specifications.

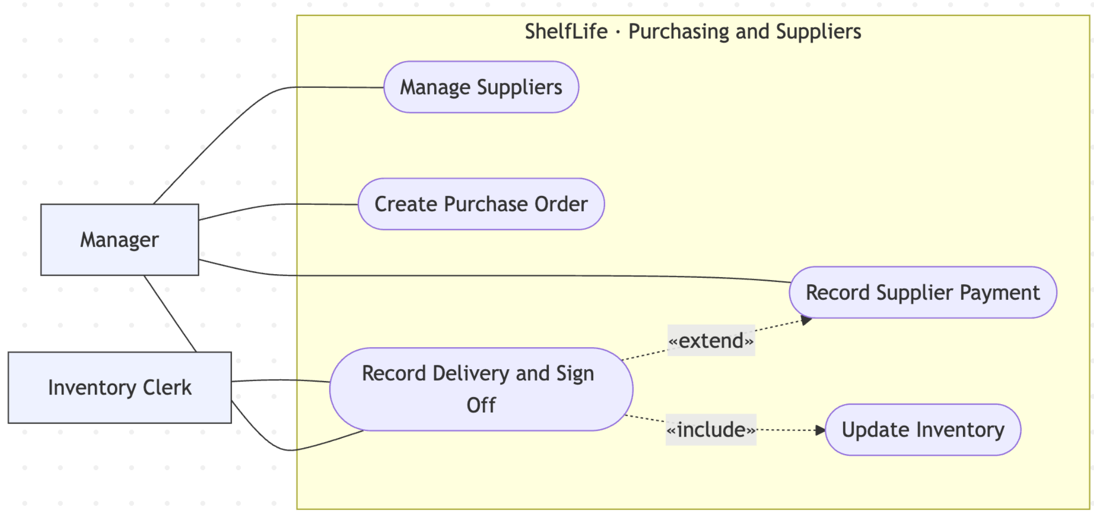
Administration and Access



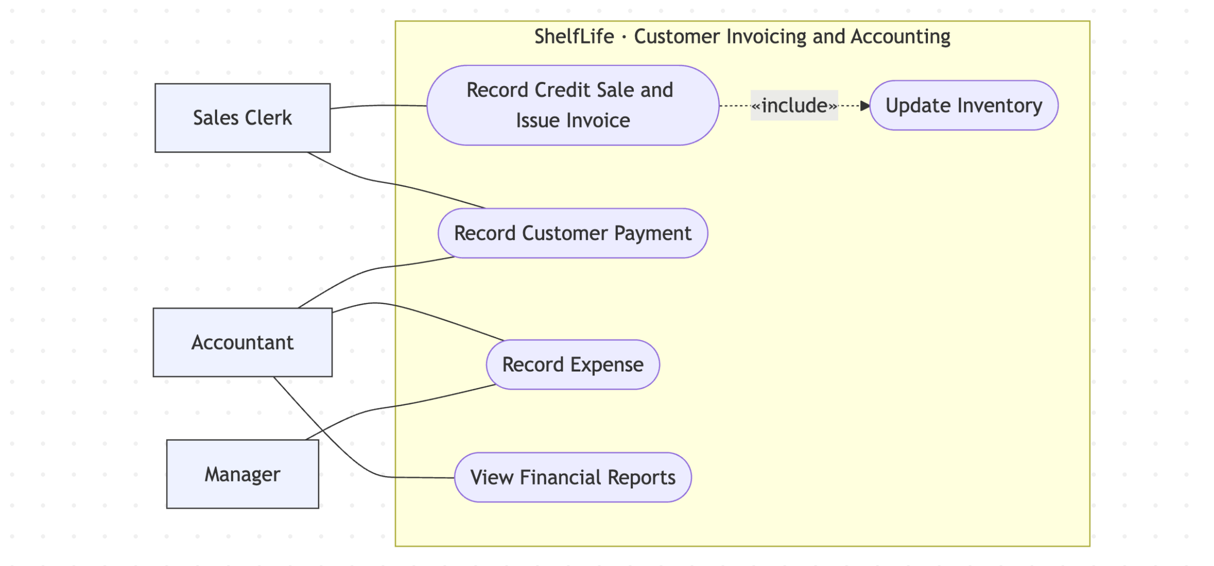
Inventory and Sales



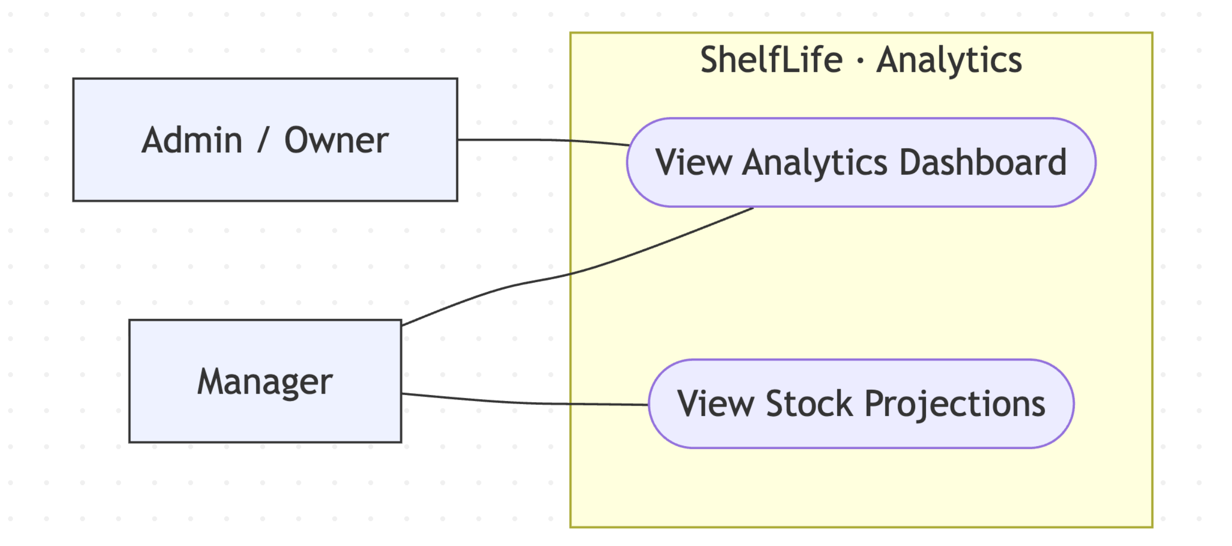
Purchasing and Suppliers



Customer Invoicing and Accounting



Analytics



3.2 Use-Case List

Use Case	Primary Actor	Subsystem
Login	Any User	Auth and Permissions
Invite Employee and Assign Role	Admin/Owner	Administration
Manage Product Categories	Admin/Manager	Administration
Manage Products and Define Unit Breakdown	Inventory Clerk	Inventory
Set Product Price	Manager	Sales
View Stock and Low-Stock Alerts	Manager	Inventory
Record Sale	Sales Clerk	Sales

Manage Suppliers	Manager	Purchasing
Create Purchase Order	Manager	Purchasing
Record Delivery and Sign Off	Inventory Clerk	Purchasing
Record Supplier Payment	Manager	Purchasing
Record Credit Sale and Issue Invoice	Sales Clerk	Customer Invoicing
Record Customer Payment	Sales Clerk/Accountant	Customer Invoicing
Record Expense	Accountant	Expenses and Accounting
View Financial Reports	Accountant	Expenses and Accounting
View Analytics Dashboard	Admin/Manager	Analytics
View Stock Projections	Manager	Analytics

Shared included use case: Update Inventory, invoked by Record Sale, Record Credit Sale, and Record Delivery and Sign Off.

4. Use-Case Specifications

4.1 Login

- **Brief Description.** Authenticates a user and starts a session scoped to their role (Admin/Owner, Manager, Sales Clerk, Inventory Clerk, or Accountant).
- **Actors.** User (any role); Accounts Database.
- **Preconditions.** The user has an active account in the system.
- **Flow of Events**

4.1 Basic Flow, successful login

User Action	System Response
The user performs an action that requires login.	The system displays a login window requesting username and password.
The user enters username and password and submits.	The system searches the Accounts Database for the credentials. When they are found, it loads the user's profile and role, records the successful login, closes the login window, and directs the user to the screen appropriate for their role.

4.2 Exceptional Flows

4.2.1 Username not found or password incorrect

User Action	System Response
The user performs an action that requires login.	The system displays a login window requesting username and password.
The user enters username and password and submits.	The system cannot find the username, or the password does not match. It displays "Unable to log in. Username or password is incorrect," with an OK button.
The user clicks OK.	The system returns to the login window for another attempt.

Post-Conditions. On success, the user has a session scoped to their role. Otherwise the system state is unchanged.

Business Rules.

- All access is authenticated, and every subsequent action is limited by the user's role (RBAC).

Nonfunctional Requirements.

- Credentials are stored and verified securely, and passwords are never displayed in plain text.

4.2 Invite Employee and Assign Role

- **Brief Description.** Adds a new employee to the store and grants them a role.
- **Actors.** Admin/Owner (or Manager); Accounts Database.
- **Preconditions.** The actor is logged in with admin or manager rights, and the store is set up.
- **Flow of Events**

Basic Flow, add an employee

User Action	System Response
The actor opens employee management and selects "invite employee."	The Login use case is invoked first if the actor has not logged in. After successful login, the system displays the employee management screen with the list of current employees.
The actor enters the employee's details and selects a role (Manager, Sales Clerk, Inventory Clerk, Accountant).	The system displays the entered details and the selected role for confirmation.
The actor confirms and saves.	The system creates the account in the Accounts Database, generates credentials, and updates the employee list to show the new employee.

Alternate Flows

1. Clerk with both duties. At Step 2 the actor selects both Sales and Inventory responsibilities, and the system records both.
2. Edit or deactivate an employee. Like the add flow: the actor selects an existing employee and chooses edit or deactivate, and the system saves the change and updates the list.
3. Exceptional Flows
4. Account already exists. If the username or email is already in use, the system reports a duplicate and does not create the account.
5. Required field missing. The system prompts for the missing field and returns to the form.

Post-Conditions. A new user account exists with the assigned role or roles and credentials.

Business Rules.

- Only Admin/Owner or Manager may assign roles.
- A clerk may be Sales, Inventory, or both.

Nonfunctional Requirements.

- Credentials are handled securely.

4.3 Manage Products and Define Unit Breakdown

Brief Description. Allows an Inventory Clerk or Manager to add, edit, or delete products, including defining the bulk-to-unit breakdown. A clerk may add and publish products but cannot edit them after publish, whereas a Manager may edit or delete.

Actors. Inventory Clerk (or Manager); Inventory Database; Rules Engine.

Preconditions. The actor is logged in with inventory rights, and a suitable category exists for a new product.

Flow of Events

Basic Flow, add a product

User Action	System Response
The clerk selects Manage Products from the menu.	The Login use case is invoked first if the user has not logged in. After successful login, the system displays the manage-products screen showing the current category and a table of its products.
The clerk selects the category for the product.	The system displays the category name and its products.
The clerk selects the "add" command.	The system brings up the Add Product screen.
The clerk enters the product information (name, SKU, bulk purchase unit and cost, sale unit) and defines the unit breakdown (for example, 1 dozen = 12 units), then clicks Save.	The Rules Engine validates the fields (SKU unique, breakdown a positive integer, cost numeric). After the rules pass, the system saves the product to the Inventory Database, closes the Add screen, and updates the manage-products table.

Alternate Flows

- Edit product (Manager). The Manager selects a product, chooses edit, changes fields, and saves. The system stores the change and updates the table. Clerks cannot edit a product after it is published (see Business Rules).
- Delete product (Manager). The Manager selects a product and chooses delete. The system removes it and updates the table.

Exceptional Flows

- Inventory Database not available.
- Improper values in fields, for example a non-numeric cost or a non-integer breakdown. The system shows a message and returns to the Add screen.
- Product or SKU already exists. The system shows a message and returns to the Add screen.
- No product selected before an edit or delete.

Post-Conditions. The add, edit, or delete is reflected in the manage-products screen and recorded in the Inventory Database.

Business Rules.

- This release.
  - The unit breakdown is a positive integer.
  - A clerk cannot edit a product after it is published, and only a Manager may edit or delete.
- Later phase. All fields are required on the Add or Edit screen, and price is expressed to two decimal places.

Nonfunctional Requirements. Works with the ShelfLife Inventory Database.

4.4 Record Sale

Brief Description. Allows a Sales Clerk to record a customer's purchase of one or more products at the prices set by the Manager. On completion, stock is decremented and per-item profit is computed.

Actors. Sales Clerk; Inventory Database; Rules Engine.

Preconditions. The Sales Clerk is logged in. The products exist and have prices set by a Manager.

Flow of Events

Basic Flow, record sale, quantity rules pass, take payment

User Action	System Response
From the main screen, the clerk begins a new sale.	The system displays the list of product categories.
The clerk selects a category.	The system displays the products in that category.
The clerk selects a product.	The system displays the product name, the Manager-set unit price, and the quantity available, and offers to add the item to the sale.
The clerk adds the product to the sale.	The system asks for the quantity requested.
The clerk enters a valid quantity and confirms.	The Rules Engine checks the quantity against available stock. After the rules pass, the system adds the item to the sale and shows the current sale, offering to add more items or to take payment.
The clerk proceeds to take payment.	The system records the sale, decrements stock in the Inventory Database (Update Inventory), computes per-item profit as (unit price minus unit cost) × quantity, and displays a completed-sale confirmation.

Alternate Flows

Quantity rules fail (invalid quantity or exceeds stock). Steps 1 to 4 of the Basic Flow take place. Then:

User Action	System Response
The clerk enters an invalid quantity (non-numeric, zero, or greater than the quantity in stock).	The Rules Engine rejects the value, and the system displays a message ("The quantity must be a positive number" or "The quantity exceeds available stock") with an OK button.
The clerk clicks OK.	The system returns to the quantity prompt.

- Customer pays on credit. At Step 6, instead of taking payment, the clerk marks the sale as credit, and control goes to the Record Credit Sale use case (which extends Record Sale).
- Add another item and continue. After Step 5, the clerk chooses to add another item, and the system returns to Step 3 (product selection).

Exceptional Flows

- Product not priced. If the selected product has no Manager-set price, the system blocks adding it and prompts that a price must be set first (Set Product Price use case).
- Inventory Database not accessible. Any step that reads or updates stock fails with a message that the inventory data is currently unavailable.

Post-Conditions. On success, the sale is recorded, stock is decremented by the sold quantities, and per-item profit is stored.

Business Rules.

- The clerk cannot change the price, and only the Manager-set price is used.
- A sale that would drive stock below zero is not allowed.
- Per-item profit = (unit price - unit cost) × quantity.

Nonfunctional Requirements. A sale can be recorded quickly, with few interactions, by counter staff. Stock updates are consistent, so a recorded sale always decrements stock.

4.5 Create Purchase Order

Brief Description. Allows a Manager to order stock from a supplier.

Actors. Manager; Inventory Database.

Preconditions. The Manager is logged in, and the supplier exists.

Flow of Events

Basic Flow, create a purchase order

User Action	System Response
The Manager selects Create Purchase Order.	The Login use case is invoked first if needed. After login, the system displays the new purchase-order screen with a supplier selector.
The Manager selects a supplier.	The system displays the supplier and an empty order-lines table.
The Manager adds products and quantities (in bulk units) and submits.	The system saves the purchase order with status "open" in the Inventory Database and displays a confirmation.

Alternate Flows

Supplier not on file. At Step 2, if the supplier does not exist, control goes to the Manage Suppliers use case, after which the Manager returns to Step 2.

Exceptional Flows

- No order lines entered. The system prompts the Manager to add at least one product line before submitting.
- Inventory Database not available.

Post-Conditions. An open purchase order is stored for the supplier.

Business Rules.

- Only a Manager may create purchase orders.

Nonfunctional Requirements.

- Works with the ShelfLife Inventory Database.

4.6 Record Delivery and Sign Off

Brief Description. Receives delivered goods against a purchase order, verifies them, signs off, updates stock, and settles payment on flexible terms (immediate, deferred, or partial).

Actors. Inventory Clerk (or Manager); Inventory Database; Rules Engine.

Preconditions. An open purchase order exists, and the goods have arrived.

Flow of Events

Basic Flow, delivery matches the order, immediate payment

User Action	System Response
The clerk selects Record Delivery and opens the matching purchase order.	The Login use case is invoked first if needed. After login, the system displays the purchase order with its expected items and quantities.
The clerk verifies the delivered items and quantities against the purchase order and enters the received quantities.	The system displays the received quantities next to the expected quantities.
The clerk signs off the delivery.	The Rules Engine confirms the received quantities are valid. After the rules pass, the system increases stock in the Inventory Database (Update Inventory), converting bulk units to sale units, and prompts for payment.
The clerk selects immediate payment and enters the amount.	The system records the payment, clears the supplier balance for this order, marks the purchase order received, and displays a confirmation.

Alternate Flows

- Deferred payment. At Step 4 the clerk chooses to defer payment. The system records the full amount as owed and updates the supplier's outstanding balance.

- Partial payment. At Step 4 the clerk enters a partial amount. The system records the payment and records the remaining amount as owed.
- Partial delivery. At Step 2 the received quantity is less than ordered. The clerk records what arrived, the system increases stock by the received quantity, and the purchase order stays partially open.

Exceptional Flows

- Quantity mismatch or discrepancy. The clerk records the discrepancy, and a Manager must confirm before sign-off.
- Inventory Database not available.

Post-Conditions. Verified stock is added to inventory, the purchase-order status is updated, and the supplier balance reflects the chosen payment.

Business Rules.

- Stock changes only upon sign-off.
- Supplier payment may be immediate, deferred, or partial.
- A clerk cannot edit a receipt after it is published.

Nonfunctional Requirements. Stock levels and supplier balances remain consistent after sign-off.

4.7 Record Credit Sale and Issue Invoice

Brief Description. Records a sale in which the customer pays later, issuing an invoice and tracking the customer's balance.

Actors. Sales Clerk; Inventory Database; Accounts Database.

Preconditions. The clerk is logged in, the customer chooses to buy on credit, and stock is available.

Flow of Events

Basic Flow, record a credit sale

User Action	System Response
The clerk records the items and quantities and marks the sale as credit.	The system displays the sale with the Manager-set prices and a customer selector.
The clerk selects or creates the customer.	The system displays the customer and the invoice total.
The clerk confirms the credit sale.	The system decrements stock in the Inventory Database (Update Inventory), issues an invoice, and increases the customer's outstanding balance in the Accounts Database.

Alternate Flows

- New customer. At Step 2 the clerk enters the customer's details, and the system creates the customer record.

Exceptional Flows

- Insufficient stock. The system blocks the credit sale of the missing quantity.

Post-Conditions. An invoice exists, stock is decremented, and the customer's outstanding balance is increased.

Business Rules. Each customer's outstanding balance is tracked, and later payments are handled by the Record Customer Payment use case.

Nonfunctional Requirements. Invoice totals and customer balances remain consistent.

4.8 Record Expense

Brief Description. Records an operating expense for financial reporting and profit calculation.

Actors. Accountant (or Manager); Accounts Database; Rules Engine.

Preconditions. The actor is logged in with accounting rights.

Flow of Events

Basic Flow, record an expense

User Action	System Response
The accountant selects Record Expense and chooses a category (rent, transport, delivery, salaries, or utilities).	The system displays the expense entry form for the chosen category.
The accountant enters the amount, date, and an optional note, then saves.	The Rules Engine validates the amount and date. After the rules pass, the system stores the expense in the Accounts Database and confirms.

Alternate Flows

- Edit or delete an expense. The accountant selects an existing expense and edits or removes it, and the system saves the change.

Exceptional Flows

- Invalid amount or date. The system shows a message and returns to the form.

Post-Conditions. The expense is stored and available to financial reports.

Business Rules

- Expenses feed financial reports and the net-profit calculation.

Nonfunctional Requirements.

- Works with the ShelfLife Accounts Database.

5. Supporting Use Cases

These use cases follow the same specification template and are summarized here.

Use Case	Primary Actor	Brief Description	Key Business Rule
Manage Product Categories	Admin/Manager	Add, edit, or delete the product categories used to organize products.	Only Admin or Manager may change categories.
Set Product Price	Manager	Set or change the selling price of a product.	Only a Manager sets prices.
View Stock and Low-Stock Alerts	Manager	View current stock and the products flagged as running low.	Low-stock is flagged from sales velocity, not a flat threshold.
Manage Suppliers	Manager	Add, edit, or delete suppliers and view their outstanding balances.	Only a Manager manages suppliers.
Record Supplier Payment	Manager	Record a later payment against a supplier's outstanding balance.	Payment reduces the supplier's balance.
Record Customer Payment	Sales Clerk/Accountant	Record a full or partial payment against a customer invoice.	Payment reduces the customer's outstanding balance.
View Financial Reports	Accountant	View revenue, COGS, expenses, and profit for a selected period.	Reports draw on recorded sales, purchases, and expenses.
View Analytics Dashboard	Admin/Manager	View sales, top products, profit, and stock status.	Visible to Admin and Manager only.
View Stock Projections	Manager	View an estimated number of days until a product runs out.	Estimated from the product's average sales rate.

6. Functional Requirements

1. The system shall reject a login with invalid credentials and grant a role-scoped session on valid credentials.
2. The system shall allow only Admin/Owner or Manager to create employee accounts and assign roles.
3. The system shall store a product with a unique SKU and reject a duplicate SKU.
4. The system shall accept a bulk-to-unit breakdown only as a positive integer and convert bulk stock into that many sale units.
5. The system shall prevent a Clerk from editing a product or receipt record after it is published.

6. The system shall display, at sale time, only the price set by a Manager and prevent a Clerk from changing it.
7. The system shall decrement stock by the sold quantity when a sale is recorded, and shall block a sale that would make stock negative.
8. The system shall compute per-item profit as (unit price - unit cost) × quantity for each sale.
9. The system shall flag a product as running low when remaining stock falls below its sales-velocity threshold.
10. The system shall allow a Manager to create a purchase order containing a supplier and one or more product and quantity lines.
11. The system shall increase stock only after a delivery is signed off against its purchase order.
12. The system shall allow supplier payment to be recorded as immediate, deferred, or partial, and shall update the supplier's outstanding balance accordingly.
13. The system shall record a credit sale as a customer invoice and increase that customer's outstanding balance.
14. The system shall record full or partial customer payments and reduce the customer's outstanding balance accordingly.
15. The system shall record an expense with a category, amount, and date, and include it in financial reports.
16. The system shall produce a financial report of revenue, COGS, expenses, and profit for a selected period.
17. The system shall present analytics for sales, top products, profit, and stock status to Admin/Owner and Manager.
18. The system shall estimate the number of days until stockout for a product from its average sales rate.

7. Nonfunctional Requirements

Category	Requirement
Security	All access is authenticated, RBAC is enforced on every subsystem, and published records are immutable to Clerks, preserving an audit trail.
Usability	Counter staff can record a sale with minimal training and few interactions.
Performance	Routine actions (record a sale, look up stock) respond quickly under normal store load.
Reliability and Integrity	Stock changes only through recorded sales and signed-off deliveries, and balances stay consistent.
Availability	The system is available during store operating hours.
Portability	The system runs in a modern web browser on desktop, tablet, or phone.
Maintainability	The system is organized into cohesive subsystems with clear responsibilities, which supports later OO design.